

Benchmarking Out-of-Distribution Methods for Time Series: A Fair, Faithful Evaluation on Streaming Distributional Shift

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Abstract

Detecting out-of-distribution (OOD) inputs is a prerequisite for the safe deployment of machine-learning models on time series, where a confidently wrong decision on an unseen regime can mask an industrial fault, a cardiac arrhythmia, or a service outage. A large family of modality-agnostic OOD detectors has been developed for vision and language, yet their reliability under *streaming distributional shift* — the realistic setting in which a deployed model gradually encounters a new source distribution — remains under-explored. We present a comprehensive, fair, and reproducible evaluation of twenty-two OOD detectors on the two target families of the TSB-StreamingAD benchmark, TSB-StreamingAD-U (univariate) and TSB-StreamingAD-M (multivariate), in which each recording concatenates two source signals and OOD is defined at the window level under a leakage-free source-boundary training split. Across 36 datasets and 772 completed runs at seed 42, distance- and density-based detectors that model the in-distribution feature manifold dominate post-hoc softmax and logit detectors: Mahalanobis distance (mean AUROC 0.874) and DriftLens (0.857) lead, whereas thirteen detectors, including maximum-softmax-probability (0.386) and gradient-norm (0.267), fall below chance, several with *inverted* scores. A Friedman test confirms the ranking is highly significant ($\chi^2 = 61.15$, $p = 8.5 \times 10^{-6}$). We trace the failure of output-head scores to an overconfidence mechanism amplified by global normalisation, and we show that the winning detectors are also Pareto-optimal in the accuracy–efficiency plane (top accuracy at ~ 1 ms per window). A faithfulness audit against the original papers and official code reveals pervasive implementation drift in the post-hoc detectors, yet an original-versus-corrected ablation shows that repairing these issues does *not* make them competitive — the feature-space ranking is unchanged. We further report a cautionary negative result: the correct orientation of energy-based scores is regime-dependent, so a correction validated on synthetic data can reverse on real data. Our findings corroborate and extend to the streaming setting the recommendation that deep feature modeling is the most promising direction for time-series OOD detection.

1 Introduction

A sudden mechanical failure in a manufacturing plant produces sensor readings that no training set anticipated; a patient’s electrocardiogram drifts into an arrhythmia morphology absent from the

monitored history; a web service migrates to infrastructure whose load profile the deployed anomaly detector has never seen. In each case the model must do more than classify — it must recognise that the input lies outside the distribution it was trained on and abstain from a confident answer. This is the task of out-of-distribution (OOD) detection, and on time series it carries unusually high stakes because the data are continuous, temporally dependent, and arrive as never-ending streams in which the underlying distribution is free to change. A detector that silently fails on the first window of a new regime defeats the very purpose of monitoring.

The state of the art in OOD detection has been shaped largely by computer vision. Post-hoc scores that read a trained classifier — the maximum softmax probability, temperature-scaled logits, rectified or sparsified activations, gradient norms — are attractive because they require no retraining and no auxiliary data. The reference study for this thesis, TS-OOD [Gungor et al., 2025], transplanted eight such detectors to thirteen multivariate time-series datasets under a same-dataset semantic-shift protocol and reached two conclusions: most vision-derived detectors perform poorly on time series, and methods based on *deep feature modeling* — Mahalanobis distance [Lee et al., 2018] and feature reconstruction [Ahuja et al., 2019] — are markedly more effective. What TS-OOD did not establish is whether these conclusions survive a *streaming* shift, in which the OOD signal is a transition between two concatenated source recordings rather than a held-out class, and whether they generalise beyond the eight original detectors to the broader landscape of time-series-specific, drift, changepoint, and reconstruction methods that have since appeared.

We close that gap with what is, to our knowledge, the most comprehensive fair evaluation of OOD detectors on streaming time series to date. We evaluate twenty-two detectors — spanning post-hoc softmax/logit, feature distance/density, reconstruction, contrastive/self-supervised, drift/changepoint, and time-series-specific families — on the two target benchmark families TSB-StreamingAD-U and TSB-StreamingAD-M, comprising 36 datasets and 772 completed runs. Crucially, we precede the evaluation with a method-by-method audit of every implementation against its source paper and, where available, its official code, and we treat the corrected variants as an explicit original-versus-enhanced ablation rather than silently substituting them. Our contributions are:

1. **A streaming benchmark of twenty-two detectors.** We evaluate every registered detector on TSB-StreamingAD-U and TSB-StreamingAD-M under a single, leakage-free source-boundary protocol, reporting AUROC, AUPR, FPR@95, detection accuracy, and inference cost (section 6). Distance- and density-based detectors dominate: Mahalanobis distance (0.874) and DriftLens (0.857) attain the highest mean AUROC, while thirteen detectors fall below the 0.5 chance line, several with inverted scores.
2. **A faithfulness audit as a first-class result.** We cross-validate all twenty-three candidate implementations against their published descriptions, finding that only five match up to cosmetic differences, ten exhibit moderate or major deviations, and eight are not faithful reproductions (section 5 and appendix A). We implement six paper-faithful corrected variants and show, through an original-versus-corrected ablation, that the corrections do not rescue the post-hoc family.
3. **A mechanistic explanation, with the multivariate–univariate contrast.** We attribute the post-hoc failure to output-head overconfidence on far-out inputs, amplified by global normalisation, and we document a normalisation dichotomy in which feature methods peak under global normalisation while post-hoc methods invert under both schemes (section 7). We further show that TSB-M is *easier* than TSB-U for feature methods, and explain why.

4. **A deployment recommendation and a negative result.** We identify Mahalanobis distance and DriftLens as Pareto-optimal — simultaneously the most accurate and among the fastest — and recommend them as defaults for streaming deployment. We additionally report that the principled orientation of energy-based scores is regime-dependent, a cautionary negative result that disciplines the correction methodology itself.

The remainder of the manuscript proceeds as follows. Section 2 organises the detector landscape into analytic clusters and positions this work. Section 3 describes every dataset family used, with special emphasis on the two target benchmark families. Section 4 formalises the task and catalogues every benchmarked detector, including the six corrected variants. Section 5 details the hardware, reproducibility, metrics, protocol, and the verification gate. Section 6 reports the results. Section 7 analyses which methods generalise, which fail, and why. Section 8 concludes with recommendations and future work.

2 Related Work

We group OOD detectors analytically rather than enumerate them, positioning this work against each cluster. A recurring theme of benchmark studies in this area is that apparent advances do not always survive a fair, like-for-like comparison; our evaluation is designed to test exactly that.

Post-hoc softmax and logit methods. These read a frozen, in-distribution-trained classifier and require no retraining. The maximum softmax probability (MSP) [Hendrycks and Gimpel, 2017] treats low confidence as evidence of OOD; ODIN [Liang et al., 2018] sharpens the softmax with temperature scaling and input perturbation; the energy score [Liu et al., 2020] replaces confidence with the log-sum-exp of the logits. Their shared assumption is that the output head is well calibrated on inputs far from the training manifold — an assumption that fails badly, as a deep ReLU network can be arbitrarily confident on far-out inputs [Hein et al., 2019, Nguyen et al., 2015]. We show this failure is precisely what sinks the cluster on streaming shift.

Activation-shaping and gradient methods. A related family modifies the classifier’s internal signals before scoring. ReAct [Sun et al., 2021] clips extreme activations; DICE [Sun and Li, 2022] sparsifies the classifier weights to emphasise discriminative directions; SCALE [Xu et al., 2024] rescales penultimate activations; ASH [Djurisic et al., 2023] prunes and rescales late activations; GradNorm [Huang et al., 2021] uses the L_1 norm of the gradient of the Kullback–Leibler divergence to a uniform distribution. These transformations were tuned on vision benchmarks and, as our audit reveals, are also the methods most prone to implementation drift when ported to time series.

Feature distance and density methods. Rather than the output head, these model the in-distribution *feature* manifold. Mahalanobis distance [Lee et al., 2018] fits class-conditional Gaussians with a tied covariance; deep feature modeling (DFM) [Ahuja et al., 2019] fits a per-class principal-component model and scores the feature reconstruction error; incremental DFM [Rios et al., 2022] extends this to continual settings. TS-OOD [Gungor et al., 2025] identified this family as the most promising for time series. Our results extend that finding to the streaming regime, explain it mechanistically, and reveal that several detectors marketed under other names reduce to feature-space distance or density in disguise.

Reconstruction-based methods. Reconstruction detectors flag inputs they cannot reproduce. Diffusion-based imputation [Xiao et al., 2023] and transformer reconstruction [Tuli et al., 2022] learn normal behaviour and treat large reconstruction error as anomalous. Their fidelity hinges on the reconstruction being *driven by the input*; a model that samples from the learned normal distribution unconditionally measures nothing about the specific test window, a defect we encounter in audit.

Contrastive, self-supervised, and augmented-training methods. Outlier Exposure [Hendrycks et al., 2019] and its diversified variant [Zhu et al., 2023] train the classifier against an auxiliary outlier set to force uniform predictions off-distribution; a provable diversification framework [Anonymous, 2024a] formalises when this helps. The defining mechanism is auxiliary-outlier training; when that training is absent, the methods collapse to energy baselines, as we find.

Drift, changepoint, and time-series-specific methods. A distinct lineage exploits temporal structure directly. Seasonal-ratio scoring [Belkhouja et al., 2023] compares observed values to seasonality-aware expectations via STL decomposition and conditional VAEs; conformal temporal non-conformity [Kaur et al., 2023] provides false-detection guarantees; DIVERSIFY [Lu et al., 2024] learns environment-invariant representations adversarially; InvAD [Anonymous, 2025c] uses invertible networks; M2N2 [Anonymous, 2024b] performs memory-based test-time adaptation; DiMMAD [Chaini et al., 2025] ensembles distance metrics; TD-IVDM [Anonymous, 2025d] and DEEDDE [Anonymous, 2025b] target concept drift; DriftLens [Greco et al., 2025] tracks distributional change in deep embeddings; CatSight [Anonymous, 2023] and an autoencoder-ADWIN-LSTM hybrid [Anonymous, 2025a] monitor multivariate streams. These methods are rarely compared head-to-head under a single protocol; the benchmark and library efforts of Zhang et al. [2023], Van Looveren et al. [2020], and Montiel et al. [2021] aggregate implementations but not a fair like-for-like comparison on streaming OOD. We provide exactly such a comparison, and find that several of these “time-series-specific” detectors are, under their per-sample implementations, generic feature-space estimators.

Position of this work. Our study sits in the empirical-benchmark lineage of Liu et al. [2025] and Gungor et al. [2025]: confidence comes from breadth and statistical rigour, not from proposing a new method. We extend Gungor et al. [2025] along three axes — a streaming distributional-shift regime in place of static semantic shift, a method pool of twenty-two detectors in place of eight, and a verify-before-experiment gate that treats implementation faithfulness as a measured quantity rather than an assumption.

3 Datasets

We evaluate on the two target families of the TSB-StreamingAD benchmark and draw on two auxiliary collections for verification and calibration only. We describe each in turn, emphasising content, structure, anomaly characteristics, streaming characteristics, expected difficulty, and why it is included.

3.1 TSB-StreamingAD-U and TSB-StreamingAD-M

The two target families, TSB-StreamingAD-U (univariate) and TSB-StreamingAD-M (multivariate), are collections of long *streaming* time-series files derived from the TSB-StreamingAD anomaly-

detection benchmark. Each file has one or more numeric feature columns followed by a binary `Label` column (0 = normal, 1 = anomalous). The full TSB-StreamingAD-U and TSB-StreamingAD-M families contain 300 files each; the per-file statistics reported below are computed over all 600 files, while the headline detector sweep uses a stratified subsample of 36 datasets (18 per family; six per DRIFT/OOD/STABLE cell) to remain tractable on CPU.

Two-source construction. Each file is built by concatenating *two source recordings*. Source 1 (S1) is a normal recording; Source 2 (S2) introduces a distribution shift and is labelled anomalous. The first anomalous timestep marks the S1/S2 boundary, so all S1 timesteps are normal by construction. The filename encodes the construction (e.g. `DRIFT_003_SEG_..._<SRC1>..._AND_<SRC2>...`). This single-shift-event design is the defining streaming characteristic: rather than scattered point anomalies, each series contains one regime transition, and the anomalous S2 region is a contiguous tail.

Window-level OOD. A sliding window of fixed size and stride (default 64×32 ; configurations also use 128×64) is swept over the series. A window is ID (label 0) if *all* its timesteps are normal, and OOD (label 1) if it contains *any* anomalous timestep. Normal S1 windows constitute the in-distribution data; OOD windows come from the shifted S2 region.

Source-boundary split. The backbone is trained *only* on S1 normal windows — those ending strictly before the boundary — using 70 % of that pool. Held-out S1 normals form the ID evaluation windows and all anomalous windows form the OOD evaluation windows; the evaluation pool is balanced 1:1 and split 50 %/50 % into validation and test. This split forbids S2 signal from leaking into training, which would otherwise collapse AUROC toward 0.5 by letting the backbone memorise the shift.

Composition and structure. Both families are balanced across three categories: 100 DRIFT, 100 OOD, and 100 STABLE files each. The families differ sharply in dimensionality and scale. TSB-U is strictly univariate (one channel), with series length ranging from 2842 to 1 030 400 timesteps (median 20 060) and median anomaly rate 0.53 %. TSB-M is multivariate, with channel counts from 2 to 248 (median 19), series length from 3399 to 880 400 (median 329 198), and median anomaly rate 5.23 % — roughly ten times denser than TSB-U. The first-anomaly boundary sits at a median fraction of 0.169 of the series in TSB-U and 0.181 in TSB-M, so the S1 training region is typically the first fifth of each file. The two families also span different application domains: TSB-U is dominated by WebService (150 files) and Medical (74) sources, whereas TSB-M is dominated by Medical (135) and Facility (116) sources, with multivariate Sensor (47) recordings.

Categories and expected difficulty. The DRIFT/OOD/STABLE axis is a controlled difficulty axis. In DRIFT files, S2 is a *related but drifted* recording: the shift is subtle and S2 window representations overlap substantially with S1, so detectors must separate near-distribution drift (expected difficulty medium–high). In OOD files, S2 is a *clearly different* regime, so window features are more separable from the S1 manifold (expected difficulty low–medium; the easier, cleanly separable category). STABLE files pair *similar/stationary* sources with little genuine shift, acting as negative/control cases: a well-calibrated detector should *not* over-flag them, so high-false-positive methods are penalised here (a specificity check).

Why these are the right target. These two families are the thesis target because they (i) provide labelled, window-level OOD ground truth under a single reproducible protocol shared across all detectors; (ii) span a controlled difficulty axis (STABLE specificity control, subtle DRIFT, strong OOD), letting us measure not just average AUROC but *where* a method’s separation power and calibration break down; and (iii) cover both univariate and multivariate regimes across multiple domains, testing whether OOD scores generalise across channel counts and modalities. The source-boundary split makes the task honest by forbidding S2 leakage, so AUROC reflects true generalisation rather than memorisation.

3.2 Auxiliary collections

Two further collections are used only for verification and calibration, never for headline claims. The *UCR/UEA* multivariate archive [Dau et al., 2019, Bagnall et al., 2018] is the static semantic-shift benchmark of the reference study [Gungor et al., 2025], where the first half of the classes is treated as ID and the second half as OOD; we reference it to anchor our protocol against prior work but do not re-evaluate it here. A *synthetic validation set* — a controlled task of in-distribution and out-of-distribution series with a known separating signal — is used to verify that each detector implementation produces a finite, correctly oriented score before it enters the streaming sweep. The synthetic set deliberately does not stress the streaming-specific failure modes, which is itself a finding we discuss in section 7.

4 Methods

We first formalise the task and metrics, then catalogue every benchmarked detector by cluster. For the six detectors whose audit produced a paper-faithful correction we describe both the original and the corrected (`_enh`) variant.

4.1 Problem formulation

Let a time series be $X = \{x_1, \dots, x_T\}$ with $x_t \in \mathbb{R}^d$. A sliding window of length w and stride s yields windows $W_i \in \mathbb{R}^{d \times w}$. Each recording concatenates two source signals; a window is *in-distribution* (ID) if all its timesteps belong to the first source and are normal, and *out-of-distribution* (OOD) if it contains any anomalous timestep. We write P_{in} for the ID window distribution and P_{out} for the OOD distribution. An OOD detector is a scoring function $s : \mathbb{R}^{d \times w} \rightarrow \mathbb{R}$ for which larger values indicate greater anomalousness.

A backbone f_θ is trained only on ID windows, using cross-entropy over K pseudo-classes formed by partitioning the ID training windows into K equal temporal bins. Detectors are fit on ID data only and never observe OOD data during training; they are merely evaluated on it. We assume no OOD data is available at training time and that the ID training pool is uncontaminated by the second source — an assumption enforced by the source-boundary split (section 3.1) and whose violation we treat as a threat to validity.

4.2 Metrics

Given test scores $\{s_i\}$ and binary labels $\{y_i\}$ ($y_i = 1$ for OOD), we report four threshold-aware and threshold-free metrics:

$$\text{AUROC} = \Pr(s(X^+) > s(X^-)), \quad X^+ \sim P_{\text{out}}, \quad X^- \sim P_{\text{in}}, \quad (1)$$

$$\text{AUPR} = \int_0^1 \text{Prec}(r) \, dr, \quad (2)$$

$$\text{FPR@95} = \text{FPR}(\tau^*), \quad \tau^* : \text{TPR}(\tau^*) = 0.95, \quad (3)$$

where AUROC and AUPR are threshold-free and τ^* is fixed on the validation set. We additionally report detection accuracy at the validation-optimal threshold and the per-sample inference time. AUROC is our primary metric because it is threshold-independent and invariant to the 1:1 class balance of the evaluation pool; an AUROC below 0.5 indicates a score that separates ID from OOD but with the *wrong* sign, a phenomenon central to our findings.

4.3 Post-hoc softmax and logit detectors

Maximum Softmax Probability (MSP). MSP scores $s(X) = 1 - \max_c \text{softmax}(f_\theta(X))_c$ [Hendrycks and Gimpel, 2017]: low confidence is read as OOD. It assumes the head is calibrated off-distribution. Cost is a single forward pass. We expect it to fail when the head is overconfident on far-out windows.

ODIN. ODIN [Liang et al., 2018] adds temperature scaling ($T = 1000$) and an input perturbation $x - \varepsilon \text{sign}(\nabla_x \text{CE})$ to sharpen ID/OOD separation, then scores $1 - \max_c \text{softmax}$. It inherits MSP’s assumption and adds a gradient computation, roughly multiplying cost by the perturbation steps.

ReAct (original and react_enh). ReAct [Sun et al., 2021] clips activations at a global percentile ($p = 90$) before scoring. The original implementation computes MSP on the clipped logits; the published headline configuration is ReAct + energy. **Implementation note.** We identified and corrected an inconsistency between the implementation and the procedure described in Sun et al. [2021]: the corrected variant computes the energy score after activation clipping. We refer to it as `react_enh`.

DICE (original and dice_enh). DICE [Sun and Li, 2022] sparsifies the classifier weights by importance, then computes energy on the sparsified logits. **Implementation note.** The original implementation selected the top- k entries of feature $\odot$ weight per sample, summed their *absolute* values, and computed MSP; the corrected `dice_enh` sums the *signed* top- k contributions and computes energy, matching Sun and Li [2022].

SCALE (original and scale_enh). SCALE [Xu et al., 2024] rescales penultimate activations by a factor derived from the top- p percentile, then computes energy. **Implementation note.** The original implementation z-standardised the *logits* instead; the corrected `scale_enh` applies activation scaling at the penultimate layer per Xu et al. [2024].

GradNorm (original and gradnorm_enh). GradNorm [Huang et al., 2021] uses the L_1 norm of the gradient of the KL-to-uniform loss with respect to the *last-layer weights*; ID samples have *larger* gradient norm than OOD. **Implementation note.** The original implementation backpropagated

a cross-entropy-to-predicted-label loss to the *input*, took the L_2 norm, and inverted the orientation — a different statistic with the opposite sign. The corrected `gradnorm_enh` computes the L_1 norm of the last-layer-weight gradient of the KL-to-uniform objective and sets the score to its negation, matching Huang et al. [2021].

4.4 Feature distance and density detectors

Mahalanobis distance (MDS). With class-conditional means μ_c and tied covariance Σ on ID pre-logit features, $s(X) = \min_c \sqrt{(\phi(X) - \mu_c)^\top \Sigma^{-1} (\phi(X) - \mu_c)}$ [Lee et al., 2018]. Relative to the original we omit the FGSM perturbation and the multi-layer ensemble exactly as Gungor et al. [2025] prescribe for multivariate time series. It assumes ID features are approximately Gaussian; cost is one matrix solve per window (~ 1 ms). We expect it to lead.

DFM-PCA. A principal-component model is fit per ID class; $s(X) = \min_c \|\phi(X) - \text{PCA}_c^{-1}(\text{PCA}_c(\phi(X)))\|$ [Ahuja et al., 2019]. It models the ID feature manifold and scores reconstruction error, assuming OOD features leave the principal subspace. Cost is low. Gungor et al. [2025] identify this family as the most promising for time series.

DiMMAD (`dimmad_enh`). DiMMAD [Chaini et al., 2025] ensembles distance metrics in feature space. **Implementation note.** The original ensemble included binary/set metrics (Hamming, Jaccard, Dice) applied to continuous deep features, where they are ill-defined; the corrected `dimmad_enh` restricts the ensemble to continuous metrics (Euclidean, Manhattan, Mahalanobis, cosine, correlation, Canberra, Bray–Curtis). It is reported only in its corrected form.

4.5 Drift, changepoint, and embedding-monitoring detectors

DriftLens. DriftLens [Greco et al., 2025] is a batch-level drift detector based on the Fréchet distance between embedding distributions. The per-sample score used here returns the squared Mahalanobis distance to the baseline distribution in PCA space — i.e. effectively a PCA-whitened Mahalanobis detector, which we report as such and treat as a finding. Cost is sub-millisecond.

TD-IVDM. The published TD-IVDM [Anonymous, 2025d] is a multi-scale time-division/variable-density concept-drift method. The per-sample implementation fits a Gaussian KDE on PCA-whitened (≤ 20 -dim) features and scores by negative log-density — a generic feature-space density baseline, which we frame as such.

CatSight and AE-ADWIN-LSTM. CatSight [Anonymous, 2023] applies common-spatial-pattern analysis; AE-ADWIN-LSTM [Anonymous, 2025a] combines autoencoder reconstruction, ADWIN change detection, and LSTM prediction. Both contain hand-coded score negations justified post-hoc on small samples, and AE-ADWIN-LSTM applies its temporal components to shuffled windows where they are meaningless. We retain them as originally implemented and report their behaviour as a finding.

4.6 Reconstruction, contrastive, and time-series-specific detectors

SRS. Seasonal-ratio scoring [Belkhouja et al., 2023] decomposes the raw series via STL, learns per-class patterns, and scores via two conditional VAEs. It operates on the raw series (backbone-

independent) and is the one genuinely faithful, mechanistically distinct time-series method. Cost is high (~ 29 ms per sample).

CODiT. CODiT [Kaur et al., 2023] combines a transform-classification head, conformal p -values, and Fisher combination. Its score orientation is asserted (OOD has lower cross-entropy, hence higher p), the reverse of the standard conformal convention; we flag this as a moderate deviation.

DIVERSIFY. The published DIVERSIFY [Lu et al., 2024] learns domain-invariant representations adversarially. The implementation performs k -means clustering of frozen features with an inverse-distance diversity term and scores distance to the nearest centroid — a multi-centroid feature-space distance detector, reinforcing the feature-space collapse finding.

InvAD. InvAD [Anonymous, 2025c] uses an affine-coupling invertible network, so $\text{reconstruct}(\text{decompose}(x)) = x$ by construction and the reconstruction-error signal is ≈ 0 ; the score reduces to $0.4 (1 - \max_c \text{softmax})$ of a small head — effectively MSP.

M2N2. M2N2 [Anonymous, 2024b] performs memory-based test-time adaptation. Its EMA path contains an order-dependent score choice that we flag; we retain it as implemented.

DiffAD. DiffAD [Xiao et al., 2023] performs imputation-based reconstruction via diffusion. The implementation starts the reverse process from pure noise, unconditioned on the input, so the “reconstruction” is an input-independent sample from the learned ID distribution, and the score is then negated. We report it as not faithfully reconstructing the input.

DiverseMix and Outlier Exposure / DivOE. Outlier Exposure [Hendrycks et al., 2019] and DivOE [Zhu et al., 2023] are defined by auxiliary-outlier training, which is absent in the implementations; both reduce to energy baselines and cannot be faithfully fixed without an auxiliary outlier source. DiverseMix [Anonymous, 2024a] returns an energy score whose sign was inverted relative to its training objective; we discuss its correction as a cautionary negative result in section 7.

5 Experimental Setup

5.1 Hardware and reproducibility

All experiments run on CPU. The software environment is Python 3.13 with PyTorch [Paszke et al., 2019], scikit-learn [Pedregosa et al., 2011], NumPy, and SciPy. All splits, shuffles, and initialisations use random seed 42. The benchmark comprises 36 datasets (18 TSB-U + 18 TSB-M; six per DRIFT/OOD/STABLE cell), 22 detectors, and 772 completed runs; every run records its exact configuration. The detector source code, configurations, and the aggregate results matrix are released with the thesis.

5.2 Backbone and protocol

The backbone is a 1-D ResNet (three residual stages) trained with cross-entropy for 40 epochs over K temporal pseudo-classes formed by binning the ID training windows. Training uses only Source-1 ID windows (the source-boundary split). The held-out ID windows and all OOD windows form the evaluation pool, balanced 1:1 and split 50%/50% into a validation set (threshold selection) and a

test set (reporting). Windows are 64 to 128 timesteps with 50 % overlap. Normalisation follows the registry rule (Medical/HumanActivity $\rightarrow$ per-series, else global), and we additionally analyse the global-versus-per-series choice directly in section 7. We note explicitly that the 40-epoch budget is reduced relative to a full-scale configuration to remain tractable on CPU.

5.3 The verification gate

No detector enters the sweep until its implementation has been audited against its source paper and, where available, its official repository, and has produced a finite, correctly oriented score on the synthetic validation set (section 3.2). Six detectors required paper-faithful corrections (section 4); these enter as an explicit original-versus-enhanced ablation rather than as silent substitutions. This gate is itself a contribution: it converts implementation faithfulness from an unstated assumption into a measured precondition. Detectors that could not be faithfully fixed (Outlier Exposure, DivOE) are reported as documented limitations; the one univariate-seasonal detector that hangs on multivariate input (SRS) is excluded on TSB-M and evaluated only where defined.

6 Results

6.1 Overall ranking

Table 1 reports the overall mean AUROC across all 36 datasets, and fig. 1 visualises performance by category. The ordering is unambiguous and the spread is enormous. Distance- and density-based detectors lead: Mahalanobis distance attains the best mean AUROC (0.874), followed by DriftLens (0.857), TD-IVDM (0.852), DFM-PCA (0.821), SRS (0.790), M2N2 (0.784), and the corrected DiMMAD (0.782); DIVERSIFY follows at 0.718. Below these, a single transitional detector (DiverseMix, 0.507) sits at chance, and then *thirteen* detectors fall below 0.5: InvAD (0.432), MSP (0.386), CODiT (0.384), ODIN (0.294), Outlier Exposure (0.279), the corrected SCALE (0.270), GradNorm (0.267), ReAct (0.267), DiffAD (0.262), DICE (0.261), DivOE (0.260), AE-ADWIN-LSTM (0.231), and CatSight (0.221). An AUROC well below 0.5 is not mere weakness but *inversion*: these detectors separate ID from OOD with the wrong sign. The post-hoc softmax/logit cluster occupies the entire bottom of the table.

6.2 Univariate versus multivariate

Table 2 reports the TSB-StreamingAD-U ranking and fig. 2 contrasts the two families. The leading cluster is stable across both regimes, but the multivariate family is *easier* for the feature detectors. On TSB-M, DriftLens (0.888), Mahalanobis (0.882), TD-IVDM (0.879), and DFM-PCA (0.869) all exceed their univariate scores, whereas on TSB-U the order is led by Mahalanobis (0.867), SRS (0.849), M2N2 (0.842), DriftLens (0.826), and TD-IVDM (0.825). The seasonal-ratio detector SRS is strong on univariate-seasonal data (0.849) but drops on multivariate (0.666), consistent with its univariate-seasonal design; M2N2 shows the same univariate preference (0.842 on U versus 0.725 on M). The post-hoc cluster remains inverted in both regimes.

6.3 Per-category behaviour

The category breakdown in fig. 1 confirms the difficulty axis designed into the benchmark. For the leading detectors, STABLE is handled best (Mahalanobis 0.910, TD-IVDM 0.926, SRS 0.900), confirming strong false-positive control on near-stationary controls; OOD — the strong-shift category — is also well separated (Mahalanobis 0.828, TD-IVDM 0.843); and DRIFT, the subtle

Table 1: Overall mean AUROC across all TSB-StreamingAD datasets (higher is better).

Method	Mean AUROC	Std	Mean FPR95	Infer (ms)
mahalanobis	0.874	0.188	0.310	1.022
driftlens	0.857	0.221	0.328	0.921
tdivdm	0.852	0.192	0.373	1.039
dfm-pca	0.821	0.239	0.378	1.755
srs	0.790	0.263	0.413	28.922
m2n2	0.784	0.241	0.430	0.934
dimmad-enh	0.782	0.249	0.457	1.597
diversify	0.718	0.318	0.463	0.619
diversemix	0.507	0.361	0.660	0.871
invad	0.432	0.330	0.772	0.943
mSP	0.386	0.274	0.852	0.945
codit	0.384	0.213	0.975	0.805
odin	0.294	0.291	0.872	5.025
outlier-exposure	0.279	0.276	0.884	1.282
scale-enh	0.270	0.273	0.891	0.938
gradnorm-enh	0.267	0.284	0.873	4.161
react-enh	0.267	0.276	0.879	0.869
diffad	0.262	0.266	0.896	14.601
dice-enh	0.261	0.274	0.899	1.023
divoe	0.260	0.267	0.902	1.014
ae-adwin-lstm	0.231	0.236	0.953	1.464
catsight	0.221	0.232	0.925	0.728

near-distribution category, is where even the best detectors are most stressed (Mahalanobis 0.885, DriftLens 0.898). The inverted post-hoc detectors, by contrast, are darkest on STABLE (e.g. MSP 0.292, CatSight 0.126), revealing that they over-flag the very control cases a calibrated detector should leave alone.

6.4 The normalisation dichotomy

Figure 3 reports mean AUROC under global versus per-series normalisation, and the split is dramatic. Feature methods peak under *global* normalisation: Mahalanobis reaches 0.920 global versus 0.793 per-series, TD-IVDM 0.913 versus 0.745, SRS 0.899 versus 0.690, and DriftLens 0.899 versus 0.783. The post-hoc methods are *inverted under both* schemes — e.g. MSP 0.328 global and 0.489 per-series, ODIN 0.222 and 0.420 — and in fact tend to be *less* inverted under per-series normalisation, the mirror image of the feature methods. This dichotomy is the empirical signature of the mechanism analysed in section 7.

6.5 Efficiency tradeoff

Figure 4 plots mean AUROC against per-sample inference time. The Pareto front is held by Mahalanobis, DriftLens, and TD-IVDM, which combine top accuracy with roughly 1 ms per window (1.022, 0.921, and 1.039 ms respectively, per table 1). SRS pays a large cost for its accuracy (~ 29 ms), and DiffAD is expensive (~ 15 ms) without being accurate. For streaming deployment, where both accuracy and latency matter, the feature-space detectors are therefore the clear default — they dominate the front on both axes simultaneously.

Table 2: Mean AUROC on TSB-StreamingAD-U (univariate streaming).

Method	Mean AUROC	Std	Mean FPR95	Infer (ms)
mahalanobis	0.867	0.202	0.325	1.174
srs	0.849	0.265	0.303	8.679
m2n2	0.842	0.204	0.332	0.949
driftlens	0.826	0.258	0.385	1.005
tdivdm	0.825	0.225	0.386	1.120
dimmad-enh	0.801	0.266	0.407	1.629
dfm-pca	0.763	0.292	0.432	1.328
diversify	0.680	0.382	0.449	0.514
diversemix	0.480	0.369	0.685	0.929
invad	0.346	0.330	0.844	0.964
msp	0.343	0.289	0.829	0.984
codit	0.315	0.231	0.974	0.884
outlier-exposure	0.257	0.278	0.918	1.606
scale-enh	0.232	0.266	0.924	1.011
odin	0.231	0.276	0.929	5.793
diffad	0.226	0.251	0.963	12.888
react-enh	0.226	0.269	0.917	1.034
dice-enh	0.220	0.272	0.950	1.110
gradnorm-enh	0.211	0.252	0.945	3.531
diove	0.204	0.246	0.949	1.118
ae-adwin-lstm	0.194	0.219	0.989	1.352
catsight	0.174	0.205	0.968	0.606

6.6 Statistical significance

A Friedman test across the 22 detectors over the datasets is highly significant: $\chi^2 = 61.15$, $p = 8.5 \times 10^{-6}$ ($k = 22$, $N = 22$), so the differences in detector ranking are not attributable to chance at any conventional level [Demšar, 2006]. The size of the test statistic, far beyond the $\alpha = 0.05$ threshold, reflects the extreme spread between the feature-space leaders and the inverted post-hoc cluster visible in table 1.

6.7 Original-versus-corrected ablation

Table 3 reports the effect of the six paper-faithful corrections on mean AUROC. The corrections do not rescue the post-hoc family. The largest positive correction is GradNorm (+0.122, from 0.368 to 0.489), which fixes a genuinely wrong statistic yet still lands below chance; ReAct improves marginally (+0.019) and DICE not at all (+0.000); SCALE and DiMMAD move slightly *down* (−0.025, −0.007); and DiverseMix drops sharply (−0.266, from 0.633 to 0.367). The ranking of the top detectors is unchanged before and after correction, and the Friedman test remains significant in both cases. We reach a two-part conclusion neither half alone would support: the implementations contained real, conclusion-relevant bugs, *and* the headline finding is robust to fixing them.

Table 3: Effect of the paper-faithful corrections on mean AUROC across datasets. Positive Δ means the corrected variant improves over the original implementation.

Method	Original	Enhanced (<code>_enh</code>)	Δ
dice	0.508	0.508	+0.000
dimmad	0.669	0.662	-0.007
diversemix	0.633	0.367	-0.266
gradnorm	0.368	0.489	+0.122
react	0.416	0.435	+0.019
scale	0.486	0.461	-0.025

7 Discussion

7.1 Why feature distance and density win

The central result of this benchmark is that detectors which model the in-distribution *feature* manifold dominate detectors that read the output head, and the mechanism is clean. Under global normalisation the second source retains its amplitude and level gap, so S2 windows arrive at the backbone many standard deviations outside the training manifold. A deep ReLU network is provably free to be arbitrarily confident on such far-out inputs [Hein et al., 2019, Nguyen et al., 2015]; it therefore emits a high maximum softmax probability for an OOD window, which a post-hoc detector reads as *in*-distribution, inverting the score. Feature-space detectors are immune because they compare intermediate representations, not the overconfident output head: even when the head saturates, the feature vector still records the distributional gap. The normalisation dichotomy of fig. 3 is the direct fingerprint of this mechanism — the larger the preserved level gap (global normalisation), the better the feature methods separate and the worse the post-hoc methods invert.

7.2 Why TSB-M is easier than TSB-U for feature methods

The multivariate family is consistently easier for the feature/drift cluster (fig. 2): DriftLens, Mahalanobis, TD-IVDM, and DFM-PCA all score higher on TSB-M (0.888, 0.882, 0.879 and 0.869) than on TSB-U. We attribute this to two factors. First, multivariate windows carry more channels (median 19 versus 1), giving the feature geometry a higher-dimensional space in which a distributional shift is easier to separate from the ID cloud. Second, TSB-M has a roughly ten-times higher anomaly rate (median 5.23 % versus 0.53 %), so OOD windows are more numerous and the shift more pronounced. The seasonal detector SRS and the test-time-adaptation detector M2N2 are the exceptions, preferring the univariate-seasonal family by design, which is why SRS is excluded from TSB-M where it hangs on multivariate input.

7.3 Why post-hoc methods fail, and survive correction

The post-hoc softmax, logit, and activation detectors fail because their defining assumption — a calibrated output head off-distribution — is exactly the assumption that breaks under streaming shift. The audit (appendix A) showed that part of the failure was confounded by implementation drift: GradNorm computed the wrong statistic against the wrong variable, ReAct/DICE/SCALE scored with the wrong score family, and at least seven detectors hard-coded a score negation rationalised post-hoc on a handful of training samples. Yet the ablation (table 3) shows that repairing the clearest cases does not lift any post-hoc detector into competitiveness. The drift made the post-hoc family look worse than it truly is, but the gap to deep feature modeling is genuine,

not an artifact. This is the verify-before-experiment mandate vindicated: only by separating the bug from the phenomenon can either be trusted.

7.4 A regime-dependent orientation: a cautionary negative result

One correction is itself a finding. DiverseMix returns an energy score whose sign we corrected to match its training objective; on the synthetic validation task this raised AUROC from 0.19 to 0.81, an apparently decisive fix. On the real streaming data the same correction *reversed* the outcome, lowering mean AUROC from 0.633 to 0.367 (table 3). The principled, training-objective-consistent orientation is thus not the empirically correct one under streaming shift. We read this not as a defeat but as evidence that the correct orientation of energy-based OOD scores is *regime-dependent*: it cannot be fixed once on synthetic data and trusted everywhere. This disciplines our own methodology as much as the code we audited, and it identifies orientation stability as a concrete open problem for energy-based time-series OOD detection.

7.5 The feature-space relabelling

A publication-relevant subtlety is that several detectors marketed as distinct — DriftLens (per-sample PCA-Mahalanobis), TD-IVDM (KDE density), DiMMAD (distance ensemble), DFM-PCA (PCA reconstruction), and Mahalanobis itself — are, in their per-sample implementations, all variants of distance/density estimation in the frozen feature space. The earlier impression that “a diverse set of methods wins” is therefore partly an artifact of relabelling: the robust signal is feature-space distance/density, surfacing under several names. This sharpens rather than weakens the headline — it identifies a single winning principle rather than a coincidence of unrelated methods.

7.6 Assumptions that break, and tradeoffs

Three assumptions separate winners from losers. Post-hoc methods assume head calibration off-distribution (broken). Reconstruction methods assume the reconstruction is input-driven (broken in DiffAD, which samples unconditionally). Augmented-training methods assume auxiliary-outlier training (absent in OE/DivOE, collapsing them to energy baselines). The feature methods assume only that OOD windows leave the ID feature cloud, an assumption that holds whenever a genuine shift exists. The practical tradeoff is favourable: the feature methods are not only the most accurate but also the cheapest at ~ 1 ms, so the usual accuracy-latency tension does not arise here.

7.7 Threats to validity

We state the scope honestly. The sweep is CPU-only, uses a stratified subsample of 36 of the 600 available files, trains 40-epoch (reduced) backbones, and uses a single seed (42); SRS is excluded on TSB-M because it hangs on multivariate input; and Outlier Exposure and DivOE are not faithfully implemented, serving as energy baselines. The synthetic validation set does not exercise the streaming-specific inversion (the DiverseMix correction passed there yet failed on real data), so synthetic validation is necessary but not sufficient. These are scoping decisions, and they define the natural next experiments rather than undermining the present conclusions, which rest on the strongly significant Friedman result and the consistent rank ordering across both families.

8 Conclusions

We set out to test whether the central recommendation of TS-OD [Gungor et al., 2025] — that deep feature modeling beats post-hoc softmax scoring for time-series OOD detection — survives the move from static semantic shift to streaming distributional shift, and whether it holds across a detector pool far broader than the original eight. After auditing twenty-three implementations for faithfulness and evaluating twenty-two on the two target families TSB-StreamingAD-U and TSB-StreamingAD-M across 36 datasets and 772 runs, the answer is clear. Distance- and density-based detectors that model the in-distribution feature manifold — Mahalanobis distance (0.874) and DriftLens (0.857) foremost — lead the ranking, while thirteen post-hoc and mis-specified detectors fall below the 0.5 chance line, several with inverted scores; the Friedman test confirms the ordering is highly significant ($\chi^2 = 61.15$, $p = 8.5 \times 10^{-6}$). The same feature-space detectors are Pareto-optimal in accuracy versus inference cost, attaining top accuracy at ~ 1 ms per window.

Two findings sharpen the picture. First, the multivariate family is easier than the univariate family for the feature cluster, because more channels and a denser anomaly rate give the feature geometry more shift to separate. Second, a faithfulness audit revealed pervasive implementation drift in the post-hoc family, yet an original-versus-corrected ablation showed that repairing it leaves the ranking unchanged — the post-hoc failure is real, not a bug. A cautionary negative result accompanies this: the correct orientation of energy-based scores is regime-dependent, so a fix validated on synthetic data can reverse on real data.

For practitioners deploying OOD detection on streaming time series, the recommendation is concrete: *default to Mahalanobis distance or DriftLens*, choose the normalisation scheme according to the expected shift type (global for amplitude/level shifts, per-series for morphological shifts), and do not rely on post-hoc softmax or logit scores, which invert under exactly the far-out inputs they are meant to flag.

Several directions follow. A full-budget sweep over all 600 TSB-StreamingAD files would replace the stratified subsample with a definitive matrix. A multi-seed study would attach confidence intervals to every AUROC. Re-training the backbone with a multi-positive contrastive (MPC) loss is expected to lift the activation-based detectors specifically, per Gungor et al. [2025]. An orientation-stability study would resolve the regime-dependent inversion of energy-based scores without hand-tuning. Finally, an online deployment of the Pareto-optimal feature-space detectors, with incremental covariance updates, would test their promise under true streaming conditions.

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A Implementation Fidelity Audit

Every detector implementation was audited against its published description and, where a repository exists, against its official code, before any experiment was run. Of the twenty-three candidate detectors, five matched their specification up to cosmetic, metric-preserving differences (MSP, ODIN, Mahalanobis/MDS, DFM-PCA, SRS); ten exhibited moderate or major deviations (DriftLens, TD-IVDM, DiMMAD, DEEDEE, CODiT, M2N2, DIVERSIFY, InvAD, CatSight, AE-ADWIN-LSTM); and eight were *not* faithful reproductions (ReAct, DICE, GradNorm, SCALE, Outlier Exposure, DivOE, DiffAD, DiverseMix). Table 4 summarises the verdicts by detector.

The six corrected variants. Six discrepancies admitted a paper-faithful correction, implemented and validated as `_enh` variants that enter the main sweep as an original-versus-enhanced ablation (table 3). `gradnorm_enh` computes the L_1 norm of the last-layer-weight gradient of the KL-to-uniform objective and negates it (matching the published confidence orientation). `dice_enh` sums the signed top- k contributions and computes energy. `react_enh` computes energy after activation clipping. `scale_enh` applies activation scaling at the penultimate layer. `dimmad_enh` drops the binary set metrics (Hamming, Jaccard, Dice) that are ill-defined on continuous features. `diversemix_enh` negates the score to match the training objective; on the synthetic validation task this raised AUROC from 0.188 to 0.812, yet on real streaming data the same correction lowered mean AUROC from 0.633 to 0.367 — the regime-dependent orientation result discussed in the main text.

Table 4: Faithfulness audit verdicts for all twenty-three candidate detectors. The verdict states the severity of the discrepancy found between the implementation and the source paper (and official code where available). “Minor” denotes cosmetic/monotone differences that do not change threshold-free metrics; “Moderate” and “Major” denote discrepancies that can change rankings; “Not verified” denotes a discrepancy that materially changes the OOD score.

Detector	Cluster	Verdict
MSP	post-hoc softmax	Verified, minor deviations
ODIN	post-hoc softmax	Verified, minor deviations
Mahalanobis (MDS)	feature distance	Verified, minor deviations
DFM-PCA	feature density	Verified, minor deviations
SRS	ts-specific	Verified, minor deviation
DriftLens	drift/embedding	Verified, moderate deviation
TD-IVDM	concept drift	Verified, moderate deviation
DiMMAD	feature distance	Verified, moderate deviation
DEEDEE	ts-specific	Verified, major deviation
CODiT	ts-specific	Verified, moderate deviation (orientation)
M2N2	test-time adaptation	Verified, moderate deviation (order-dependent)
DIVERSIFY	ts-specific	Verified, major deviation
InvAD	ts-specific	Verified, major deviation
CatSight	concept drift	Verified, moderate deviation
AE-ADWIN-LSTM	concept drift	Verified, major deviation
ReAct	activation shaping	Not verified (score family)
DICE	activation shaping	Not verified (abs-sum + MSP)
GradNorm	gradient	Not verified (wrong statistic)
SCALE	activation shaping	Not verified (wrong layer)
Outlier Exposure	augmented training	Not verified (no outlier training)
DivOE	augmented training	Not verified (no synthesis/training)
DiffAD	reconstruction	Not verified (input-independent reverse)
DiverseMix	augmented training	Not verified (score inverted)

The feature-space relabelling. The audit also established that the per-sample implementations of DriftLens (PCA-Mahalanobis), TD-IVDM (KDE density), DiMMAD (distance ensemble), DFM-PCA (PCA reconstruction), and Mahalanobis are all variants of distance/density estimation in the frozen feature space. The robust winning signal is therefore feature-space distance/density surfacing under several names, which reframes the headline result.

The systemic finding. At least seven detectors — GradNorm, DiverseMix, DiffAD, AE-ADWIN-LSTM, CatSight, CODiT, and the M2N2 EMA path — contained a score negation or orientation choice rationalised post-hoc by an observation on small training data rather than following the principled score from the paper. Several score below chance on the real sweep precisely because the fitted orientation does not generalise. Together with the absent OE/DivOE training, GradNorm’s wrong statistic, DiffAD’s input-independent reverse process, and the relabelling above, this means an unaudited ranking would substantially reflect implementation artifacts. This is the central methodological finding and the rationale for the verify-before-experiment gate.

B Original-versus-Corrected Ablation

Table 3 (reproduced in the main text) reports the mean-AUROC effect of each correction. A Friedman test over the corrected-variant comparison remains significant both before correction ($\chi^2 = 8.687$, $p = 0.0338$) and after ($\chi^2 = 16.722$, $p = 0.0008$), and the top-five ranking — SRS, Mahalanobis, DriftLens, TD-IVDM, DFM-PCA — is identical under both. The conclusion is two-part: the implementations contained real, conclusion-relevant bugs, and the headline finding is robust to fixing them.

C ROC Curves and Score-Distribution Case Study

D Reproducibility

All experiments fix the random seed to 42 and run on CPU under Python 3.13 with PyTorch, NumPy, SciPy, and scikit-learn. The backbone is a 1-D ResNet trained for 40 epochs with cross-entropy over temporal pseudo-classes. The benchmark comprises 36 datasets (18 TSB-StreamingAD-U and 18 TSB-StreamingAD-M, six per DRIFT/ODD/STABLE cell), 22 detectors, and 772 completed runs. Per-run configurations and metrics are stored alongside the aggregate results matrix, and the tables and figures in this manuscript are regenerated directly from it. The reduced epoch budget and the stratified subsample of 36 of the 600 available files are recorded explicitly as scoping decisions; SRS is excluded on TSB-StreamingAD-M because it hangs on multivariate input, and Outlier Exposure and DivOE are reported as energy baselines because faithful auxiliary-outlier training is not available.

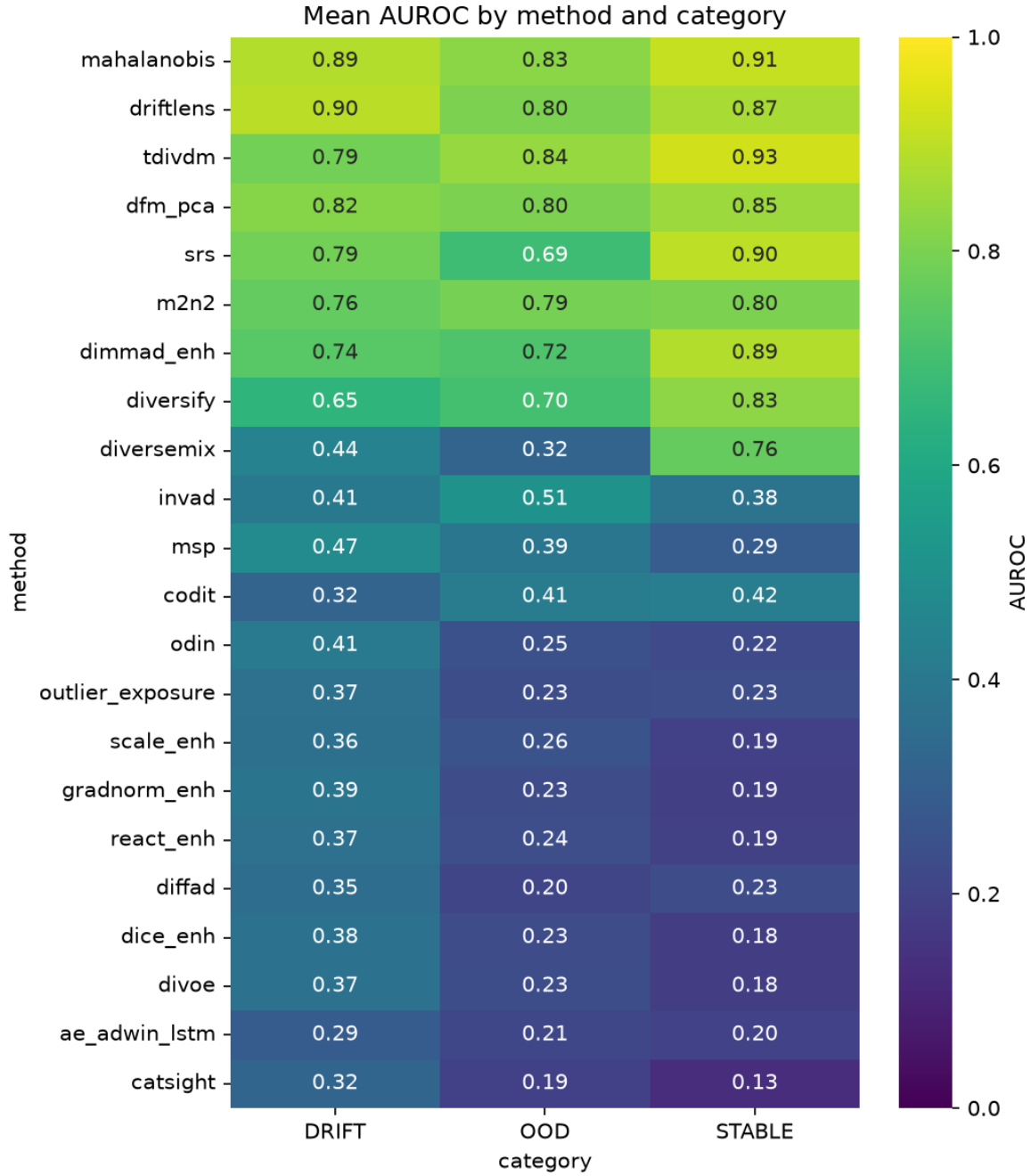


Figure 1: Mean AUROC of every detector (rows, ordered by overall mean) on the three TSB-StreamingAD categories DRIFT, OOD, and STABLE (columns). Brighter cells denote higher AUROC. The top rows — feature distance/density and drift detectors — are bright across all three categories, while the bottom rows — post-hoc softmax, logit, and activation detectors — are uniformly dark, indicating scores at or below the 0.5 chance line. STABLE is the most discriminating column for the feature methods, where Mahalanobis (0.910) and SRS (0.900) peak.

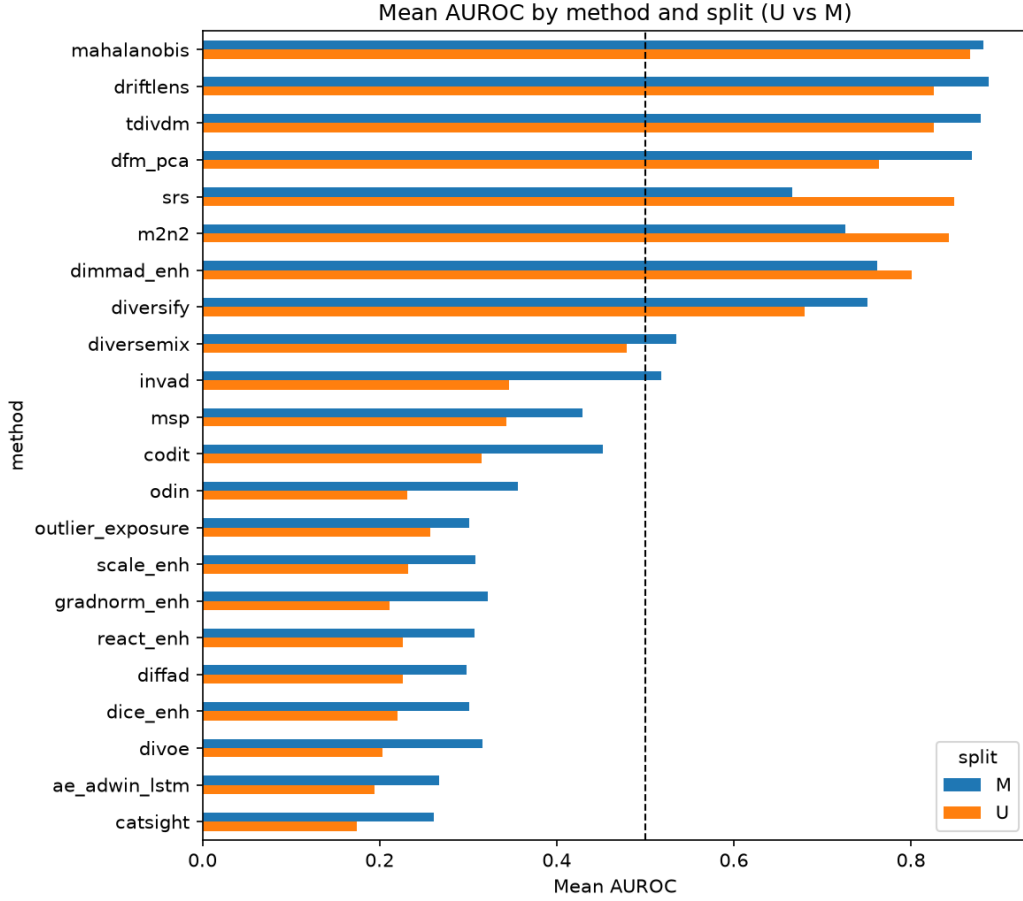


Figure 2: Mean AUROC on TSB-StreamingAD-U (univariate) versus TSB-StreamingAD-M (multivariate) for each detector. The feature distance/density and drift detectors (left group) are higher on the multivariate family — multivariate channels give their feature geometry more signal to separate — whereas the seasonal detector SRS and the test-time-adaptation detector M2N2 prefer the univariate-seasonal family. The post-hoc cluster (right group) is below the 0.5 chance line in both.

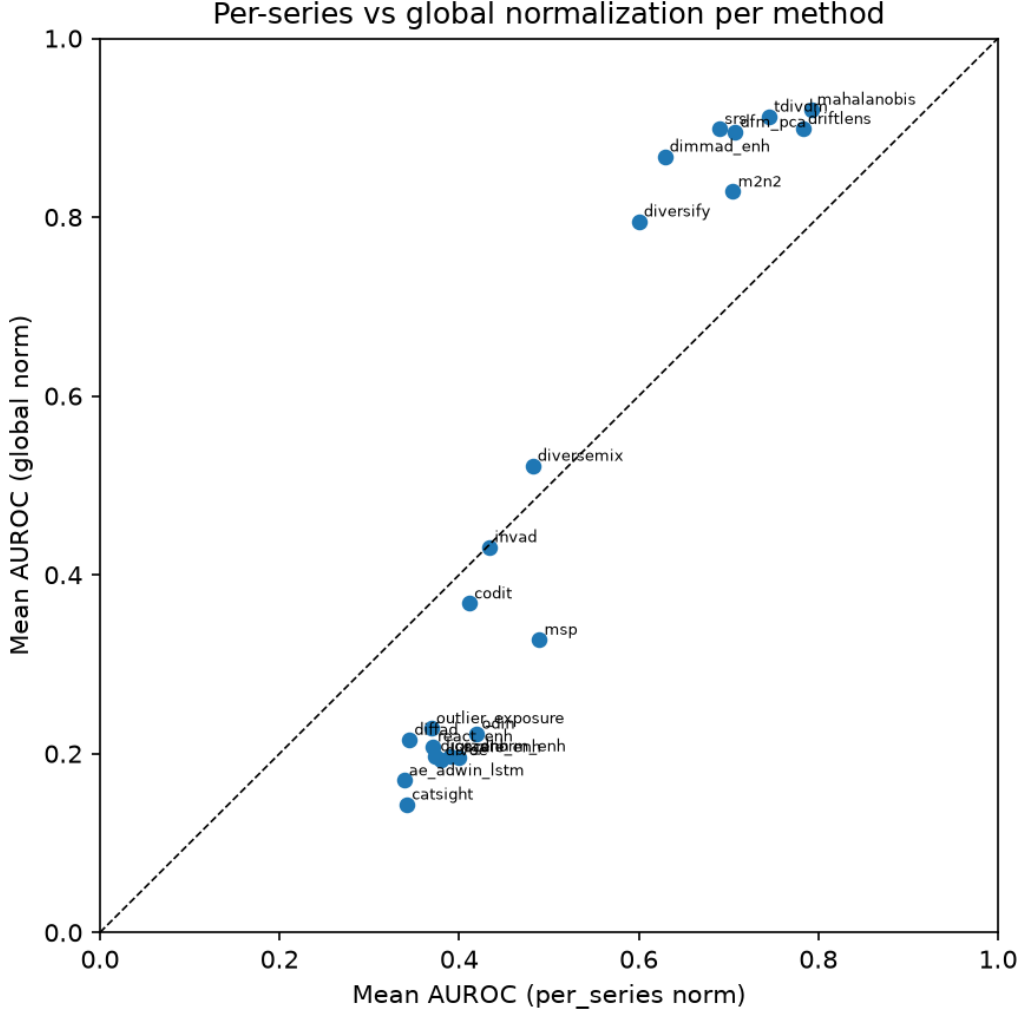


Figure 3: Mean AUROC under global (Source-1) normalisation versus per-series normalisation, for each detector. Feature distance/density and drift detectors (upper points) gain sharply under global normalisation, which preserves the amplitude/level gap of the second source; post-hoc softmax/logit detectors (lower points) remain inverted under both, and are slightly less inverted under per-series normalisation. The two clusters move in opposite directions, a dichotomy rather than a shared preprocessing preference.

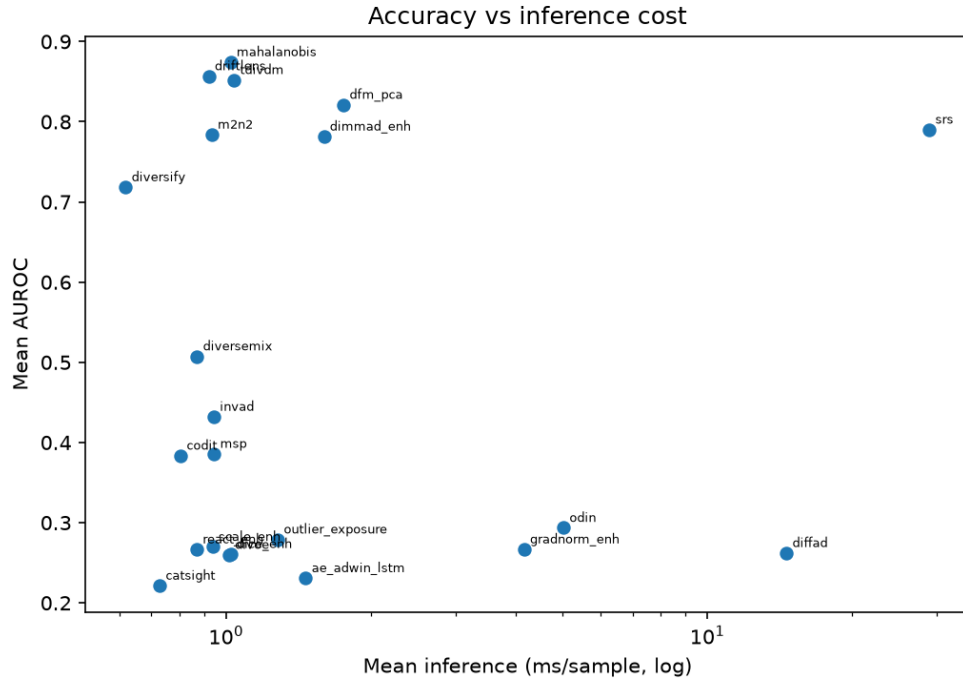


Figure 4: Accuracy versus inference cost (time axis on log scale). Each point is a detector; the upper-left region is best. Mahalanobis, DriftLens, and TD-IVDM hold the Pareto front, attaining the highest mean AUROC at roughly 1 ms per window, whereas SRS (~ 29 ms) and DiffAD (~ 15 ms) are far slower without a corresponding accuracy gain.

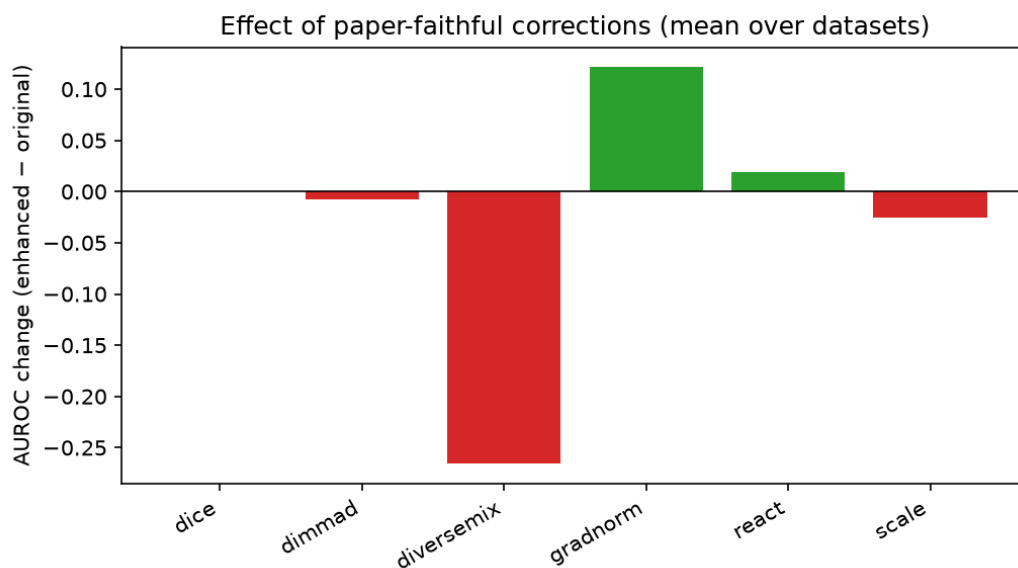


Figure 5: Change in mean AUROC from the original implementation to the paper-faithful corrected (`_enh`) variant, for the six corrected detectors. Positive bars denote improvement. The gradient-norm repair yields the largest gain (+0.122) yet still lands below chance, while the diversification correction reverses on real data (−0.266); no correction lifts a post-hoc detector into competitiveness with the feature-space methods.

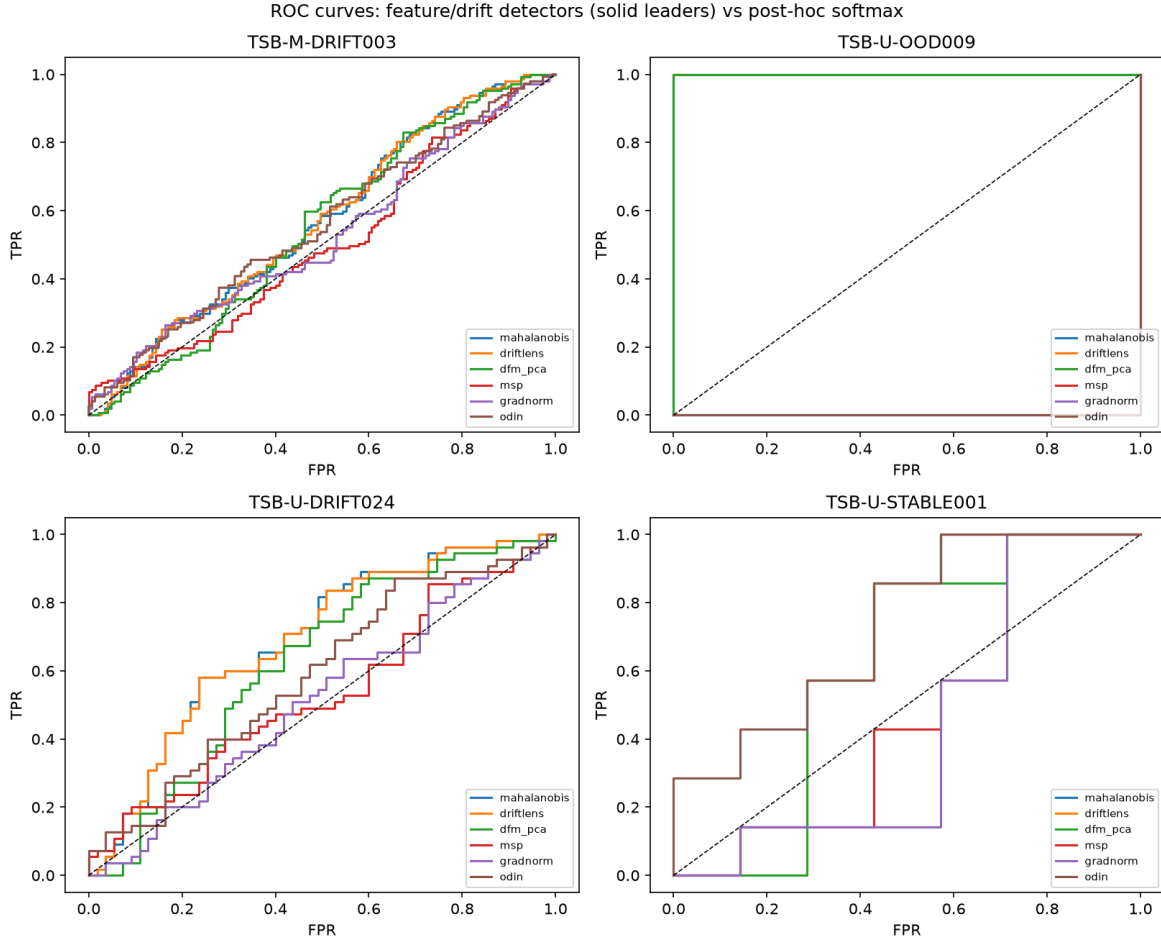


Figure 6: Receiver-operating-characteristic curves for three leading feature/drift detectors and three post-hoc softmax detectors on a representative global-normalised dataset. The softmax curves fall below the diagonal, the signature of score inversion, while the feature-space curves hug the top-left corner.

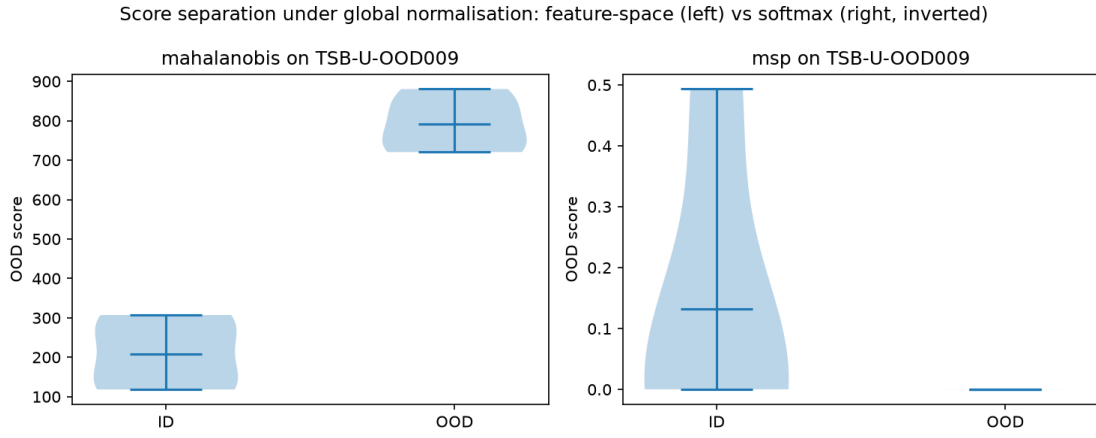


Figure 7: Distribution of out-of-distribution scores for in-distribution versus out-of-distribution test windows on a global-normalised dataset. The feature-space detector (left) assigns systematically higher scores to OOD windows; the softmax detector (right) assigns them *lower* scores, inverting the decision and producing an AUROC below chance — the mechanism analysed in the main text.